SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

15/20

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I G.K. Vyshnavi / 192425069 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

1. Network Solution for a Remote Rural Branch Office

A multinational company is planning to establish a new branch office in a remote rural location where high-speed wired internet services are unavailable. The branch must support cloud-based applications, video conferencing, and secure communication with the headquarters. At the same time, the company has a limited budget and requires reliable connectivity to avoid interruptions in business operations. Therefore, the network solution must balance bandwidth, reliability, security, cost, and future scalability.

Recommended Internet Connection

The most suitable solution would be to use a 4G/5G Fixed Wireless Access (FWA) connection as the primary internet connection, provided that reliable mobile network coverage is available in the location. A high-gain outdoor antenna can be installed if necessary to improve signal strength and connection quality.

4G/5G connectivity is more suitable than installing a new wired connection because laying fibre or other wired infrastructure in a remote rural area can be expensive and may not be practically available. A 5G connection can provide sufficient bandwidth for cloud applications, video meetings, file transfers, and normal business communication.

However, since uninterrupted operations are required, relying on only one connection would be risky. Therefore, a secondary backup connection should also be installed. This could be a connection from a different mobile network provider or a satellite internet connection if mobile coverage is unreliable. A dual-WAN router can automatically switch to the backup connection when the primary connection fails. This provides better reliability and reduces network downtime.

Recommended Networking Devices

The branch office should use the following networking devices:

- Dual-WAN business router: Connects both the primary and backup internet connections and provides automatic failover.
- Managed switches: Connect computers, printers, servers, and other wired devices while allowing VLAN configuration and traffic management.
- Wireless access points: Provide secure Wi-Fi connectivity for employees and business devices.
- Next-Generation Firewall (NGFW): Protects the network from unauthorized access, malware, and other cyber threats.
- UPS (Uninterruptible Power Supply): Provides temporary power to important networking devices during short power failures.

The dual-WAN router should also support Quality of Service (QoS). Since the available bandwidth may be limited, QoS can prioritize important traffic such as video conferencing, cloud applications, and communication with headquarters. Less important activities, such as software downloads or personal browsing, can be given lower priority. This helps maintain good network performance even when many users are connected.

Security Mechanisms

Because the branch office must communicate with the headquarters and access company resources, strong security mechanisms are necessary. A Virtual Private Network (VPN) should be established between the branch office and headquarters. The VPN encrypts data transmitted over the public internet and protects sensitive business information from unauthorized access.

A Next-Generation Firewall should be used to monitor incoming and outgoing traffic. It can block suspicious connections, restrict unauthorized applications, and apply security policies. The network should also use network segmentation through VLANs to separate different types of users and devices, such as employee systems, guest Wi-Fi, and administrative devices.

Employees accessing sensitive company resources should use Multi-Factor Authentication (MFA). This provides an additional layer of security because access requires more than just a password. Regular software updates, antivirus protection, strong password policies, and security monitoring should also be implemented.

Reliability and Cost Considerations

The proposed solution is designed to provide good reliability without exceeding the company's limited budget. Using 4G/5G or satellite connectivity avoids the high cost of installing new wired infrastructure in a remote location. Using a backup connection from another provider improves availability without requiring expensive dedicated leased lines.

Automatic failover ensures that if the primary connection becomes unavailable, the network can quickly switch to the backup connection. A UPS can also protect essential networking equipment from short power interruptions.

Future Scalability

The proposed network can be easily expanded as the branch office grows. Additional access points, switches, and users can be added without redesigning the entire network. If higher bandwidth becomes available in the future, the company can upgrade the primary internet connection while continuing to use the same network architecture.

Therefore, the recommended solution is a 4G/5G primary connection with a secondary backup connection, dual-WAN router, managed switches, wireless access points, VPN, firewall, MFA, VLANs, and QoS. This design provides a practical balance between bandwidth limitations, reliability, security, cost, and future scalability, making it suitable for a remote rural branch office.

2. Wireless Network Design for a University Campus

A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, hostels, and other common areas. The network must support a large number of simultaneous users while minimizing signal interference and providing secure access. Since the university has a limited budget and cannot deploy an unlimited number of access points, the wireless network must be designed carefully to balance coverage, performance, security, and cost.

Access Point Placement

Before deploying the network, a wireless site survey should be conducted. The survey helps identify the size and layout of buildings, walls and other physical obstacles, areas with weak signals, and locations where a large number of students are expected to connect simultaneously.

Since only a limited number of access points are available, they should be placed strategically rather than equally throughout the campus. Areas with a high concentration of users, such as libraries, lecture halls, laboratories, classrooms, and hostels, should be given higher priority. For example, a large lecture hall containing hundreds of students may require multiple access points, while smaller offices or low-traffic areas may require fewer. Access points should be mounted in central and elevated locations, such as ceilings or corridors, to provide better signal coverage.

The access points should also be connected to the university's central network using high-speed wired links. Power over Ethernet (PoE) switches can be used to provide both network connectivity and electrical power to the access points through a single Ethernet cable. This reduces installation complexity and cost.

Frequency Band Selection

The network should use dual-band or Wi-Fi 6/6E access points that support multiple frequency bands. The 5 GHz band should be preferred in high-density areas because it provides more available channels and experiences less interference than the 2.4 GHz band. The 2.4 GHz band provides better coverage over longer distances and can be used in areas where wider coverage is more important than maximum speed. However, it has fewer non-overlapping channels and is more likely to experience interference from other wireless devices.

Therefore, the network can use both bands based on the requirements of each location. Modern access points can use band steering to move capable devices to the 5 GHz band while allowing older or long-range devices to continue using 2.4 GHz. This helps distribute users and improves overall network performance.

Proper channel planning is also necessary. Nearby access points should be configured to use different non-overlapping channels to reduce interference. The transmission power of access points should also be adjusted carefully so that signals do not overlap excessively.

Managing a Large Number of Users

Since more than 3,000 students may access the network, the university should use a centrally managed wireless controller or cloud-based wireless management system. This allows administrators to monitor all access points, manage users, configure security policies, and detect network problems from a central location.

The network should also use load balancing to distribute users among nearby access points. If one access point becomes overloaded, some devices can be directed to another nearby access point. Quality of Service (QoS) can also be configured to prioritize important traffic such as online classes, educational applications, and university services over non-essential traffic.

To reduce congestion, bandwidth limits can be applied to individual users or certain types of traffic. This prevents a small number of users from consuming most of the available bandwidth through large downloads or streaming.

Authentication and Network Security

For secure access, students and staff should not be allowed to connect using a common password. Instead, the university should use WPA2-Enterprise or WPA3-Enterprise authentication with a centralized RADIUS server.

Students and staff can log in using their individual university usernames and passwords. This provides better security because access can be controlled, monitored, and revoked when necessary.

The network should also be divided into separate VLANs for students, staff, administrators, laboratories, and guests. This prevents unauthorized users from accessing sensitive university systems. A separate guest network can provide internet access without allowing access to internal resources.

Additional security measures such as firewalls, encryption, strong password policies, and regular security monitoring should also be used to protect the wireless network from unauthorized access and cyberattacks.

Balancing Coverage, Performance, Security, and Cost

The proposed design provides an effective balance between the different constraints. Instead of placing access points everywhere, they are deployed strategically in high-density and important areas, reducing the total number of devices and installation costs.

Using both 2.4 GHz and 5 GHz bands provides a balance between coverage and performance. Proper channel planning, band steering, and load balancing help minimize interference and network congestion.

Centralized management reduces the operational effort required to maintain a large wireless network, while enterprise-level authentication and VLAN segmentation provide secure access without requiring completely separate physical networks.

Therefore, the recommended solution is a centrally managed Wi-Fi network using strategically placed dual-band access points, 5 GHz for high-density areas, 2.4 GHz for wider coverage, proper channel planning, load balancing, WPA2/WPA3-Enterprise authentication, RADIUS-based login, VLAN segmentation, and QoS. This design can support a large number of students while maintaining good coverage, performance, security, cost efficiency, and future scalability.

3. Secure Network Architecture for a Financial Institution

A financial institution handles highly sensitive information such as customer details, account information, transaction records, and online banking services. Therefore, the organization must strengthen its cybersecurity while ensuring that employees can access the resources required for their work. The network design must also comply with strict security policies without significantly reducing network performance or creating excessive operational costs. A suitable solution should therefore focus on network segmentation, firewall protection, strong authentication, intrusion detection, and continuous monitoring.

Network Segmentation

The network should be divided into separate segments using Virtual Local Area Networks (VLANs) and access control policies. Instead of allowing all devices and systems to communicate freely, different departments and services should be separated.

For example, separate network segments can be created for:

- Employee workstations
- Customer information and databases
- Online banking servers
- Administrative and management systems
- Public-facing services
- Guest or visitor networks

This segmentation limits the movement of attackers within the network. If one employee device is compromised, the attacker will not automatically gain access to sensitive databases or banking systems. Access between different VLANs should be controlled using firewalls and access control rules.

A Demilitarized Zone (DMZ) should also be created for public-facing services such as online banking websites and external applications. The DMZ separates these internet-facing servers from the internal network and sensitive databases. Even if a public-facing server is attacked, the internal banking systems remain protected by additional security layers.

Firewall Deployment

A Next-Generation Firewall (NGFW) should be deployed at the boundary between the organization's internal network and the internet. The firewall should inspect incoming and outgoing traffic and block unauthorized connections, malicious traffic, and suspicious applications.

In addition to the main perimeter firewall, internal firewalls should be placed between critical network segments. For example, communication between employee networks and customer databases should pass through controlled firewall rules.

The firewall can also provide features such as:

- Application control
- Malware filtering
- Web filtering
- VPN support
- Deep packet inspection
- Access control policies

This layered firewall approach improves security without unnecessarily slowing down the entire network because strict inspection can be focused mainly on sensitive areas and high-risk traffic.

Authentication and Access Control

maintain good performance, security policies should prioritize the protection of sensitive banking systems while avoiding unnecessary inspection of low-risk internal traffic. Network monitoring can help administrators identify performance issues and security threats at the same time.

The architecture is also scalable. As the financial institution grows, additional VLANs, firewall rules, servers, and security monitoring tools can be added without completely redesigning the network.

Conclusion

Therefore, the recommended secure network architecture should use VLAN-based network segmentation, a DMZ for public-facing services, Next-Generation Firewalls, Multi-Factor Authentication, Role-Based Access Control, IDS/IPS, and centralized security monitoring. This design follows a layered security approach and provides strong protection for sensitive customer and banking information. At the same time, it maintains network performance, supports compliance requirements, controls operational costs, and allows the network to expand in the future.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately